
UNIT 16 HAZARDOUS AND E -WASTE MANAGEMENT

Structure

- 16.1 Introduction
- 16.2 Objectives
- 16.3 Hazardous Waste: Introduction
- 16.4 Classification of Hazardous Waste
- 16.5 Treatment of Hazardous Waste
 - 16.5.1 Physical treatment
 - 16.5.2 Chemical treatment
 - 16.5.3 Biological treatment
 - 16.5.4 Incineration
 - 16.5.5 Land disposal
- 16.6 Hazardous Waste Management in India
- 16.7 Case Studies
 - 16.7.1 Minamata Bay incidence
 - 16.7.2 Bhopal Gas Tragedy
 - 16.7.3 Basel Convention
- 16.8 Introduction to E-waste
- 16.9 E-Waste Issues
- 16.10 E-Waste Solutions
- 16.11 Let Us Sum Up
- 16.12 Key Words
- 16.13 References and suggested Readings
- 16.14 Answers to Check Your Progress Exercises.

16.1 INTRODUCTION

The industrial and domestic activities of humans generate huge quantities of waste. These wastes are classified as municipal, electronic, industrial, biomedical and radioactive waste etc., based on their characteristics and source of origin. Some of these wastes are hazardous by nature while the others can become hazardous on exposure to different environmental components. The hazardous waste can find its way to our homes and can cause detrimental effect to our surroundings. On the other hand, we have electronic waste that may seem completely harmless in their existential forms but can prove harmful when processed unscientifically for extraction of precious and semiprecious elements. A clear understanding of the characteristics and classification of these wastes is a must to prevent the harms it can incur to the people and environment.

16.2 OBJECTIVES

After reading this unit, you will be able to:

- Explain Hazardous Wastes, characteristics, classification and its impact on health and environment.
- Describe the processing and disposal of hazardous wastes.
- Explain the electronic waste and its characterisation.
- Discuss the concerns associated with E-waste processing and recycling.

16.3 HAZARDOUS WASTE: INTRODUCTION

Human beings depend on environment for almost all their needs. If we continue with our current rate of development, resource consumption and pollution, time is not far when we would not be left with anything for our survival. Pollution control efforts are needed at mass scale, both at international and national level. India has a history of initiatives taken to protect the environment. Laws like Water Act of 1974, the Air Act of 1981 and many others were promulgated early in the stages of development to protect the interest of people and environment.

In addition to these, a landmark act was put in place to keep a check on widespread use and contamination by hazardous substances that in due course have found a place in our food chain and water cycle (incidents of Mercury poisoning - Minamata disease (case study in Box 1), arsenic poisoning, lead contamination-affecting the functioning of brain etc.). The accident in Union Carbide Factory in Bhopal (case study in Box 2) in 1984 led to the formulation of the Environment Protection Act (EPA) of 1986 wherein, the subject of hazardous waste was dealt comprehensively for the first time. In 1989, the hazardous waste management and handling rules were notified for the first time. These rules were further amended from time to time in the year 2000 and 2003 and a new set of rules known as Hazardous waste (Management, Handling and Transboundary Movement Rules) 2008 were formed. The rules cover the responsibilities of different state and central authorities in managing hazardous waste including the Ministry of Environment, Forests and Climate Change (MOEFCC), Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs) etc. The latest set of Rules were notified in 2016 known as the Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2016 under the EPA. These rules explicitly mention the hazardous and other wastes it applies to (table 1) and shall **not** apply to other sources that are also deemed hazardous to manage like:

- a) **Effluent water and exhaust gases** as covered under the provisions of the Water Act, 1974 and the Air Act of 1981 respectively and the guidelines made thereunder and further amendments;
- b) **Shipping operations waste** beyond five kilometres of the relevant baseline as described under the provisions of the Merchant Shipping Act, 1958 and the guidelines made thereunder and additional amendments;

- c) **Radio-active wastes** as described under the Atomic Energy Act of the year 1962 provisions including any further amendments;
- d) **Bio-medical wastes** covered under the regulations of Bio-Medical Wastes Rules, 1998 and amendments; and
- e) **Municipal Solid Wastes** as described in Municipal Solid Waste Management Rules, 2000, and amendments.

16.4 CLASSIFICATION OF HAZARDOUS WASTE

Hazardous waste can thus be defined as any waste that can be dangerous to health and environment independently or after interacting with other substances or environment because of their physical, chemical and biological characteristics by virtue of its reactivity, toxicity, flammability, explosiveness and corrosivity (**Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2016**). The source of hazardous wastes can be complex. They can be generated from residential, non-point and industrial sources, but industrial processes are known to be substantial contributors to the hazardous waste generation.

The hazardous waste is thus classified on the following basis of **a)** processes from different industries generating it (Table 1), **b)** the constituents that are toxic beyond certain concentration, both classes A and B (Table 2 & Table 2.1) and **c)** any other waste possessing hazardous characteristics (Table 3).

Table 1: List of Hazardous Wastes based on the origin from industrial processes.

S. No.	Industries	Hazardous Waste
1.	Petroleum and petrochemicals Industry	
	Petroleum/Petrochemicals extraction, processing transportation and maintenance processes and pyrolytic operations	• Residue/debris from different sources. • Oil containing wastes. • Waste acid and alkalis. • Organic residues • Spent catalyst and molecular sieves
2.	Ores and metal processing industry	
	Processing of ores containing heavy metals like chromium, manganese, nickel, cadmium etc. and primary/ secondary production of zinc/lead/copper/cadmium and Aluminium. Metal treatment procedures like etching, polishing etc. Iron and steel production.	• Process residues containing heavy metals • Spent acid/alkalis • Flue gases with heavy metals • Waste containing toxic metals • Sludges from different processes • Nitrates wastes • Asbestos containing residues

	Processes involving arsenic and asbestos.	
3.	Chemicals Industry	
	Production of caustic soda, chlorine, acids, phenol, solvents, glues, resins, paints, lacquers, varnishes organic compounds and processes handling involving hazardous chemicals and wastes.	• Apparatus used during the decontamination • Wastes from wet scrubbers • Inorganic wastes • Spent solvents • Mercury in waste
4.	Fertilizers	• Arsenic/chromium containing sludge
5.	Pesticides	• Heavy metals • Spent chemicals
6.	Textiles	•
7.	Electronics	• Spent Chemicals and solvents
8.	Leather tanneries	• Wastes containing chromium
9.	Pharmaceuticals and health care	• Date expired products • Spent carbon and solvent
10.	Pulp and Paper	• Corrosive wastes • Organic halides
11.	Others	
	Processes involving decontamination of units involving hazardous wastes/incineration. Purification of flue gases/water from Common Effluent Treatment plants (CETPs)	• Residues/sludge generated in the process • Flue gases and water treatment wastes
Source: Hazardous Wastes (Management and Handling Rules, 2016.		

Table 2: List of a few waste constituents with concentration limits (Based on Toxicity factor)

Class	Constituents	Concentration in mg/l
1.	Arsenic	5.0
2.	Cadmium	1.0
3.	Chromium and/or Chromium (III) compounds	5.0
4.	Lead	5.0
5.	Manganese	10.0
6.	Mercury	0.2
7.	Selenium	1.0
8.	Silver	5.0

9.	Ammonia	50*
10.	Cyanide	20*
11.	Nitrate (as nitrate-nitrogen)	1000.0
12.	Sulphide (as H ₂ S)	5.0
13.	Benzene	0.5
14.	Benzo (a) Pyrene	0.001
15.	Bromodichloromethane	6.0
16.	Carbon tetrachloride	0.5
17.	Chloroform	6.0
18.	Cresol (ortho+ meta+ para)	200.0
19.	Dibromochloromethane	10.0
20.	Hexachlorobenzene	0.13
21.	Hexachlorobutadiene	0.5
22.	Hexachloroethane	3.0
23.	Naphthalene	5.0
24.	Nitrobenzene	2.0
25.	Pyridine	5.0
26.	Tetrachloroethylene	0.7
27.	Trichloroethylene	0.5
28.	2,4,5-TP (Silvex)	1.0
29.	2,4-Dichlorophenoxyacetic acid	10.0
30.	Alachlor	2.0
31.	Alpha HCH	0.001
32.	Atrazine	0.2
33.	Beta HCH	0.004
34.	Butachlor	12.5
35.	Chlordane	0.03
36.	Chlorpyrifos	9.0
37.	Delta HCH	0.004
38.	Endosulfan (alpha+ beta+ sulphate)	0.04
39.	Endrin	0.02
40.	Ethion	0.3
41.	Heptachlor (& its Epoxide)	0.008
42.	Isoproturon	0.9
43.	Monocrotophos	0.1
44.	Antimony	15
45.	Beryllium	0.75
46.	Chromium (VI)	5.0
47.	Cobalt	80.0
48.	Copper	25.0

49.	Molybdenum	350
50.	Nickel	20.0
51.	Thallium	7.0
52.	Vanadium	24.0
53.	Zinc	250
54.	Fluoride	180.0
55.	Dichlorodiphenyltrichloroethane (DDT), Dichlorodiphenyldichloroethylene (DDE), Dichlorodiphenyldichloroethane (DDD)	0.1
56.	Polychlorinated biphenyls	5.0
57.	Dioxin (2,3,7,8-TCDD)	0.001
Source: Hazardous Wastes (Management and Handling Rules, 2016).		

Table 16.1: List of Hazardous Wastes based on Total Threshold Limit Concentration (TTLC).

Serial No.	Constituent	Concentration in mg/kg (ppm)
1.	Asbestos	10000
2.	Total Petroleum Hydrocarbons (TPH) (C5 - C36)	5,000

Table 16.2: List of Wastes with hazardous characteristics.

Serial No.	Characteristics of Waste
1.	Flammable
2.	Corrosive
3.	Reactive or Explosive
4.	Toxic
5.	Substances or waste liable to spontaneous combustion
6.	Substances emitting flammable gases on contact with water
7.	Oxidizing
8.	Organic peroxides
9.	Poisons
10.	Infectious wastes-containing micro-organisms
11.	Liberation of toxic gases in contact with air or water
12.	Eco-toxic
13.	Capable of leading to any material which may possess any of the above-mentioned characteristics e.g. leachate.

The classification of hazardous waste is important in understanding the strength and source of generation for better handling and disposal of waste. The major industries involved in the generation of hazardous waste include mining operations, paints and dye industries, chlor-alkali plants, pharmaceuticals, petrochemicals, pesticides fertilisers and other industries (State of the environment, 2001). The waste from these industries if goes untreated into the soil and water systems, finds its way to food chain and is practically impossible to take out.

The World Health Organisation classify hazardous waste as waste having short term chronic impact and long-term environmental hazard based upon the threat it imposes on humans and environment (WHO, 1987). The source and health effects of some of the hazardous substances are given in Table 4.

Table 16.3: List of few hazardous substances, their sources impacts on health.

Hazardous Substance	Uses/source	Health Impacts
Metals		
Arsenic	Pesticides, medicines, glass	Toxins; dermatitis; muscular paralysis; damage to liver and kidney; loss of hair; Gangrene; cancer
Cadmium	Electroplating, batteries, fertilisers processing	Carcinogenic; lungs and kidney damage
Lead	Paints, pipes, storage batteries	Neurotoxic; causes headaches; irritability; mental impairment in children; damages brain, liver and kidneys
Mercury	Chlor-alkali cells, fungicides, pharmaceuticals	Nerve damage; kidney damage; fatal effects of alkyl mercury
Pesticides		
DDT	Insecticides	Cancer; damage liver, embryo; nervous depression
BHC	Insecticides	Cancer; embryo damage
Petrochemicals		
Benzene	Gasoline additive, solvents	Headaches; nausea; loss of muscle coordination; leukemia
Vinyl chloride	Plastics production	Lung and liver cancer, depresses central nervous system; suspected embryotoxin
Others		
Asbestos	Roofing/insulation/Air-conditioning roofs/plastics/fibres etc.	Carcinogenic to workers and users.
Source: McKinney et al. 2006; De and De, 2013		

With urbanisation and industrialisation, we have synthesised thousands of new chemicals and they are used for manufacturing a number of commercial products across the world. Our dependency on such chemicals have increased and we have totally internalised the use of such chemicals. These hazardous substances (Table 5) have now found a way to all the households and help in accomplishing day-to-day chores.

Table 16.4: List of hazardous substances commonly used at household.

Serial No.	Categories of hazardous substances	
1.	Cleaning	Disinfectants; drain, toilet and window cleaners; spot removers; septic tank cleaners
2.	Paint	Latex and oil-based paints; paint thinners and solvents; varnishes and lacquers; wood preservatives; artists paints and inks
3.	Gardening	Pesticides; flea powder
4.	Automotive	Petrol; used motor oil; battery acid; solvents; brake and transmission fluid
5.	General	Dry-cell batteries (mercury and cadmium); Glues and cements; acid batteries for invertors
Source: Miller, 2007.		

The hazardous waste generated from certain non-point sources and the households may get mixed with the municipal solid waste or directly get into the natural mediums of soil and water. This leaves only the industrial hazardous waste stream that can be treated as per waste management regulations in a scientific manner as the quantities produced are huge. The waste generated from industries can largely be divided into three categories:

- 1) Recoverable or recyclable waste,
- 2) Incinerable waste, and
- 3) Disposal of waste in landfill sites.

This categorisation of waste is there to prioritise the ways we handle hazardous waste (Figure 1). Our first priority should be to reduce the use of hazardous substances throughout the line of production. Secondly, a sincere attempt should be made to recycle the waste, i.e. to reuse the chemicals as much as possible. The waste then should be treated for reducing it to less harmful or non-hazardous forms before being disposed of in landfills and other forms like surface impoundments etc.

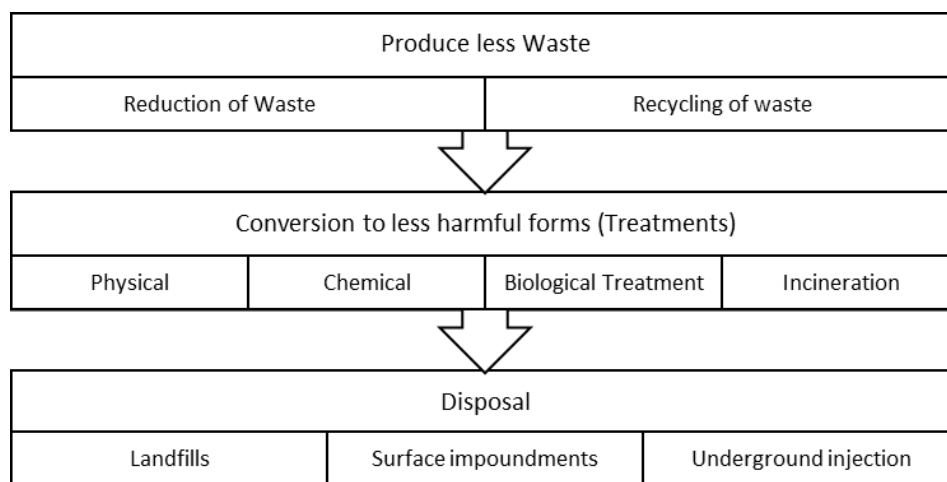


Figure 1: The series of steps for hazardous waste minimisation and treatment.

Recovery and Recycling of Hazardous Waste

The clean production in industries, through reuse and recycling are the best ways to reduce the impact of hazardous waste, as it not only increases the life of chemicals used but also reduces the consumption of raw material making the whole process more economical and environment friendly. The recycling methods adopted would vary from industry to industry with the types of chemicals used. During recycling, the waste material needs to undergo a series of treatments as discussed below:

16.4 TREATMENT OF HAZARDOUS WASTE

The waste produced from industry is treated to reduce its toxicity and hazard value and recycle some of the chemicals. This is achieved through different types of treatments discussed below.

4.5.1 Physical Treatment

- 1) **Sedimentation:** It is one of the preliminary ways to separate solid from liquid making use of the gravity. The solids separated from the liquids can be removed as sludge which can further be concentrated using evaporation, filtration and centrifugation.
- 2) **Adsorption:** It is the physical adsorption of particles to a solid surface, a system commonly used in many water purification systems. It is mostly used for the removal of organics. The surface area of the adsorbent plays an important role and a commonly used adsorbent is powder Granular Activated Carbon (GAC). Once the adsorbent is clogged in the process of cleaning, it is either replaced or regenerated. Regeneration is an expensive and an energy intensive process.
- 3) **Aeration:** For volatile chemicals, aeration can be used to take them out of solution. Air and steam can be used for this system. In the commonly used air stripper, the contaminated water is sprayed downward through a packing material in tower, while air is blown upward carrying the volatiles with it. This process can remove almost 95 % of the volatile

organic compounds (VOCs) such as trichloroethylene, tetrachloroethylene, trichloroethane, benzene, toluene and other common organics derived from solvents.

- 4) **Reverse osmosis:** It uses pressure to force the contaminated water against semi-permeable membrane. The membrane allows only water to pass through it and larger particles are filtered out.
- 5) **Ion exchange:** In this process, the ions present in waste water are exchanged with ions present in the exchange resin. It is used to remove toxic metal ions present in waste water.
- 6) **Electrodialysis:** It uses ion selective membranes and electric fields to separate cations and anions in the solution. Metal salts from plating rinses can be removed in this way.

16.5.2 Chemical Treatment

Chemical treatment of hazardous waste can help break down hazardous waste into less hazardous or relatively stable chemicals and sometimes help recover important by-products from them. This helps in reduction of raw material consumption and prolongs the use of material, imposing less burden on disposal. The chemical treatment required would depend upon type of waste in question.

- 1) **Neutralisation:** the corrosive hazardous waste can be chemically neutralised. Acidic waters can be neutralised with slaked lime ($\text{Ca}(\text{OH})_2$), whereas alkaline wastewaters can be neutralised by passing CO_2 through it which produces carbonic acid (H_2CO_3). Sometimes, simultaneous neutralisation of acidic and caustic waste is also possible.
- 2) **Chemical Precipitation:** some toxic metals can be precipitated from the solution by decreasing their solubility which is pH dependent. With proper pH adjustments toxic metals can be precipitated out from the solution. For a solution containing more than one toxic metals, the pH has to be adjusted in multiple phases.
- 3) **Chemical reduction-oxidation (redox):** an ion or atom is said to be reduced when it gains electrons and oxidised if it loses ones. Both the reaction of reduction and oxidation occur simultaneously. These reactions are used to change the valency of certain ions as they depict different reactivity at different valencies, for e.g. hexavalent chromium Cr (VI) is more toxic than trivalent chromium Cr(III). The wastes that can be treated with redox oxidation reactions include benzene, phenols, most organics, cyanide, arsenic, iron and manganese. While Cr(VI), mercury, lead silver and chlorinated organics like PCBs and unsaturated hydrocarbons can be treated using redox reduction methods.

One of the technologies involves the use of *cyclodextrin*, a type of sugar made from corn starch to remove toxic materials like solvents and pesticides from soil and ground water (Miller,2007).

16.5.3 Biological Treatment

- 1) **Bioremediation:** This involves the use of micro-organisms to decompose the hazardous waste to a less harmful state. It is considered to be an eco-friendly approach towards waste management as it does involve the use of extra chemicals. This way, the cost of treatment is significantly reduced, although the time taken is more than usual. As it involves living organisms, it becomes sensitive as the survival temperature, pH and nutrient supply have to be continuously maintained. Moreover, biological treatment of hazardous waste is one the intermediary stages of treatment. There have been many successful pilot projects but the field application at large scale is yet to succeed completely. A significant development has been made in the field of biological treatment especially those for the treatment of VOCs.
- 2) **Phytoremediation:** It is the use of plants for the removal of pollutants from soil and water. Some plants have affinity for metals and some of them can form complexes in soils hence blocking their availability. This is also a cost-effective method for removal of pollution.

16.5.4 Incineration

Heating of hazardous waste at high temperatures of the order of 2000deg C can break them down and convert them to less harmful substances. The process has its own merits and demerits. This is particularly effective with organic wastes, and many mutagens, teratogens and pathological wastes. However, it is not very effective in destroying inorganic chemicals, but they can be concentrated in ashes. Incineration is a complex process, and a lot of care needs to be taken for proper combustion of wastes, like temperature, length and time of combustion and the amount of oxygen available for combustion. All these parameters determine the emissions from an incinerator. It may emit noxious gases (carcinogens), unburnt particles of original waste, odours, carbon monoxide, hydrogen chloride, oxides of sulphur and nitrogen. The emissions are noxious even in case of municipal solid waste. As these emissions have effects on the surrounding communities, incinerator technology has question marks attached to its feasibility.

16.5.5 Land Disposal

The land disposal of hazardous waste is part of the age-old practice. Dumping of hazardous waste on land have shown its impacts in the long run. Just like any other method, it has its own downsides. Land disposal techniques include landfills, surface impoundments and deep well injections.

- 1) **Landfills:** The Central Pollution Control Board of India has specified guidelines for construction of landfill sites for hazardous waste. Some important precautions while making a landfill include that the site should be lined with thick layers of resistant material to avoid any ground water contamination through leachate. The leachate produced should be collected regularly and treated, the emission of gases from the site should be taken care of in a scientific manner and regular environmental monitoring systems should be there, around the site. Even when all the

regulations are followed the land cannot be used for any other purpose long after the site is abandoned (MoEF, 2010).

- 2) **Surface impoundments:** These are open excavated or diked areas for storage of liquid hazardous waste. They are a sort of temporary arrangement before the waste finally makes it to the landfill site. As the impoundments are open, the waste is accessible, and is used for treatment methods like precipitation, sedimentation and biodegradation. Because of its temporary nature and poor siting, they can be a source of groundwater contamination.
- 3) **Underground Injection:** It involves injection of wastewater deep under the ground where it is beyond the reach of contamination of ground water. They are generally more than 700m below the surface. Even then they have leakage problem and hence are not considered completely safe (Masters, 2001).

16.6 HAZARDOUS WASTE MANAGEMENT IN INDIA

The industries are responsible for the management of hazardous waste generated, yet the quantities are huge and sometimes beyond their capacity to manage. To facilitate the proper disposal of hazardous waste to prevent it from entering the natural environment, the government has TSDF (Transport, Storage, Disposal Facilities) in different states to collect the waste and safely incinerate and dispose them off. According to a CPCB report, there are 36,165 hazardous waste generating industries in India generating 62,32,507 metric tonnes of waste every year. Out of the total waste, 49.55% is recyclable (CPCB, 2009). The land disposable portion is 43.78 % and 6.67 % is incinerable. There are around 22 Common TSDFs in 10 states only. In total, there capacity is much less than the total land disposable waste generated. Similarly, for incinerable waste, there are 14 common incinerators in 7 states and 127 individual incinerators in 12 states. The total capacity of incinerators is also much less than the waste generated.

Table 6: List of common alternatives for household cleaners.

S.No.	Product	Possible Alternatives
1.	Drain cleaners	A combination of baking soda and vinegar
2.	Oven cleaners	A combination of dish wash liquid and borax powder and warm water
3.	Toilet cleaners	Borax powder, warm water and lemon juice
4.	Glass/ window cleaners	Furniture: Vinegar, water and wipe dry with news papers
5.	Furniture/floor polish	Mineral oil and lemon Wood floor: mineral oil and vinegar
6.	Detergents	Use less or no-phosphate detergents/laundry soaps/use borax powder in case of hard water.

Source: HAPPI, 2010.

As the data suggests, India has a lot of waste to handle, even more than the capacity. The need of the hour is to strictly implement the regulations to manage it, reduce the consumption of substances involving the generation of hazardous wastes. The primary responsibility of industries should be to go for cleaner production techniques. At household level, the people can go for more environment friendly alternatives available (Table 6).

Check your Progress Exercise 1

- 1) Make a list of all the products that we use daily at home that are listed under hazardous category or can prove hazardous when disposed.

16.7 CASE STUDIES

4.7.1 Minamata bay incidence

The Minamata bay in Japan now is widely known for its contamination with a highly toxic metal, mercury. It was first discovered in 1950s (1953 to 1960) in Japan. Many people lost their lives or were crippled for their life by feeding on fish catch from the bay. Genetic defects were observed in children whose mothers fed on the fish during pregnancy. The source of mercury in the bay was a pesticide plant. It was discovered later that the compound known as methyl mercury (CH_3Hg) was bioaccumulated in fish and caused toxicity. Interestingly, the elemental forms of mercury are mostly inert and non-toxic but the vapours when inhaled are highly toxic. But the methyl mercury is the most toxic forms of all as they are soluble in fat and can pass through cell membranes or can be deposited in fat tissues. This can cause permanent damage to brain and nervous system. The mercury released from the pesticide plant was the inorganic mercury. It was a mystery as to what caused the conversion of inorganic mercury to methyl mercury. Years of research was able to establish that the presence of an anaerobic, methane synthesising bacteria in water that caused this. This conversion occurs in presence of Co(III) containing vitamin B_{12} coenzyme. The methyl group is transferred from methyl cobalamin to mercury ion leading to formation of methyl mercury or dimethyl mercury. The conversion of dimethyl mercury to methyl mercury is facilitated in acidic medium. The impact of methyl mercury contamination is escalated because of bioaccumulation causing a lot more damage to life. The Minamata bay is restricted for fishing since then. (Kudo and Miyahara, 1991; De and De 2013). On the same lines, a thermometer factory in Kodaikanal, India was disposing the hazardous waste containing mercury on the slopes of Pambar River in a haphazard manner. The people living downstream and the factory workers were affected with neural disorders. The factory was shut down in the year 1997 after a sufficient damage was done (Toxic pathways, 2003).

16.7.2 Bhopal Gas Tragedy

The accident that took place in the Union Carbide pesticide factory in Bhopal on the night of December 3, 1984 is one of the most tragic ones in the history

of mankind. Methyl isocyanate, an intermediary product of pesticide manufacturing, was leaked, and reacted with water releasing huge fumes of the toxic gas (exothermic reaction). The gas spread over an area of more than 77square kilometres, exposing 60000 people. Almost 7000 people died and 50000 suffered from some kind of ailments (McKinney, 2007). Till that time there was no legal regulations existing to manage the situation. This led to the formulation of Environment Protection Act, 1986 a comprehensive law for the protection of environment and hazardous waste management.

16.7.3 Basel Convention

It is an international agreement on the control of movement of hazardous waste across the countries and their disposal. It was adopted on 22nd March, 1989 and was signed by 105 states with an objective to protect the environment and humans from the detrimental effects of hazardous wastes. The convention aims at the reduction of hazardous waste generation and the use of environmentally sound management techniques. It also puts a restriction on its trans boundary movement making an exception wherever it is done in an environmentally sound manner and also developing a regulatory system in place wherever the transboundary movement applies. India is also a signatory to the convention. The e-waste movement to different countries also comes under the norms of Basel convention (<http://www.basel.int/>)

E-Waste Management

16.8 INTRODUCTION: E-WASTE

Technological advancement and the need of transfer of large amount of information across the world in the shortest possible time have made us all depend on electronic gadgets. Be it television, computers, laptops, tablets, mobile phones etc., they all are innate parts of our life now. In the last 15-20 years we have seen a paradigm shift in the way we communicate and the information is processed. Switch over from TVs to plasma to LED screens, reducing size of hard disks, the use of mobile phones by almost everyone has made us dependant on them for every small thing.

There are no second thoughts, as how these electronic devices have saved on large amount of resources by processing data using computers, especially when large government sector shifts to the use of online tools. On the other hand, these fast-technological changes, mass outreach and now, the availability of cheaper internet and data services make any device obsolete in a short time. The amount of waste from those useless gadgets is piling up and needs to be taken care of systematically. The waste thus generated from any equipment using electricity or electronic field to make it functional is called as WEEE (Waste from Electrical and Electronic Equipment). This terminology was adopted by European Union in 2002 and they have streamlined the management of this type of waste to an extent (EU, 2002). Under the Environment Protection Act, 1986, the (E-waste Management Rules), 2016, e-waste is defined as any electrical or electronic equipment discarded as waste in part or in whole by a consumer or bulk consumer and

the rejects from the manufacturing, refurbishment and repair process. A list of common e-waste generated is given in Table 7.

Table 7: List of e-waste products from consumer and or bulk consumer.

S. No.	E-waste product	S. No	E-waste product
1.	Television sets (including sets based on (LCD-Liquid Crystal Display and LED-Light Emitting Diode technology)	7.	Centralised data processing: Mainframes, Minicomputers
2.	Refrigerator	8.	Printers including cartridges
3.	Washing Machine	9.	Personal Computing: Notepad Computers
4.	Air-conditioners excluding centralised air conditioning plants	10.	Personal Computing: Personal Computing: Notebook Computers
5.	Fluorescent and other Mercury containing lamps	11.	Answering systems
6.	Cellular telephones	12.	Personal Computers (Central Processing Unit with input and output devices)
7.	Personal Computing: Laptop Computers (Central Processing Unit with input and output devices)	13.	Electrical and electronic typewriters
8.	Copying equipment	14.	Facsimile
9.	User terminals and systems	15.	Telephones
10.	Telex	16.	Cordless telephones
11.	Pay telephones		
Source: E-waste Rules 2016			

16.9 E-WASTE ISSUES

The hazardous waste, (Management and Handling Rules amendments, 2003, defines that electrical and electronic equipment using any hazardous substance listed in its schedules would be considered hazardous. Later, e-waste Management Rules were promulgated in 2007. The latest set of rules of the year 2016 and their amendments in 2017 have made an attempt to look after all the aspects of e-waste management from manufacturing to disposal, defining the roles of government agencies, users (both individual and commercial) and the recyclers.

The e-waste management is a big crisis that the world faces today. The first problem is that the scale of generation is huge. A study, conducted by ASSOCHAM-NEC in 2018 showed that the world production of e-waste is expected to reach 52.2 million tonnes by 2021. India generates almost 2 million tonnes of waste and is among the top 5 contributors to e-waste. Within India, e-waste generation is highly uneven. The ten states contribute to 70 % of waste generation (Maharashtra, Tamil Nadu, Andhra Pradesh, Uttar Pradesh, West Bengal, Delhi Karnataka, Gujarat, Madhya Pradesh and Punjab) and 65 cities contribute to 60% waste (IRGSSA, 2005). The main source of this e-waste is from government, public and private sectors (70%), around 15% comes from household and the remaining from the manufacturing units. Moreover, India also receives e-waste from developed nations. Though, the transport of E-waste is regulated under the norms of Basel convention, this e-waste comes to India in the form of donations (Toxics Link, 2007), adding to the woes of managing it.

This mass scale electronics production and wastes are a huge burden on natural resources consumption. Composition wise, a large part of e-waste is made up of iron and steel, wood, plastics, some toxics metals like lead, cadmium, mercury, brominated flame retardants etc. and some valuable metals like, gold, silver copper etc. One mobile phone typically contains 15g of copper, 0.34g of silver and 0.034g of gold. The presence of these valuable metals makes recycling of e- waste, an economically lucrative option, provided it is done in a scientific manner. The CPCB, in association with state Pollution Control Boards and industries has set up recycling units in different states. The guidelines to scientifically recycle the e-waste is laid down by CPCB. Just as the case with hazardous waste, the capacity of recycling and disposal sites is far less than the rate of generation. Only a small, 5 % of the waste gets recycled in formal sector (ASSOCHAM), 2018). The remaining 95% is managed by the unorganised sectors of trash dealers.

Whenever the waste from a household or an office goes to a trash dealer, it generally lands up in an illegal recycling unit. The recycling of waste is done in a very crude manner like manually breaking and dismantling objects with sharps, burning of plastic casing of wire to get copper, extraction of precious metals using strong acids or even illegal dumping of e-waste along with municipal solid waste. These unscientific ways of recycling create air pollution (release of carcinogens) and various non-point sources of water and soil pollution. The labour works at cheaper rate, have no safety gadgets, gloves, masks etc to prevent them from exposing themselves to the toxic hazardous substances. Absence of any financial security and health insurance makes it all the more difficult. Sometimes, child labour is also involved. The reason why this sector is able to sustain and propagate itself is because of the huge money involved in it. The money enables them to connect to all the nodal points of collection making a strong network.

16.10 E-WASTE SOLUTIONS

To deal with this waste of humongous proportions, the need of the hour is to have a policy with multi stakeholders in it. The users should be encouraged

for prolonged use of their devices increasing their lifetime. Educated and informed customers can make smart choices. The European Union Association of Waste from Electrical and Electronic take back systems observed 13th October, 2018 as the first International Electronic Waste Day where India was also a part of the initiative. Being signatory of the Basel convention, and for environment protection we should monitor the import of e-waste from developed nations, coming for recycling. We need more of organised recycling units in formal sector as the other options of landfill dumping and incineration to dispose of the waste, come with their own set of problems where environment is severely affected. The Extended Producers Responsibility (EPR) is a principle to take back the used products to avoid it from getting it into illegal recycling route. This way, both, the consumer and producer participate in protecting the environment and resource conservation. Hence, there cannot be one solution to this complex problem but efforts from all directions can bring about significant changes.

Check your progress Exercise 2

- 1) What are the problems associated with processing and disposal methods of hazardous and electronic waste?

16.11 LET US SUM UP

With the modernisation and development of human society, we are becoming more and more dependent on products that require hazardous inputs for production or become hazardous over its usage. Howsoever we may use these products, the damage to the environment incurs either during production, usage and disposal. The laws in place, [Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2016 and the Electronic Waste Management Rules, 2007] describe in detail the ways of handling and disposing these wastes. Proper regulatory services and the contribution of citizens is a must to make the enforcement of these laws successful. Awareness amongst citizens about the concerns related to usage and disposal of hazardous and electronic wastes is a must to protect the environment from degrading. A conscious use of these products is required to ensure sustainable use of natural resources that are exploited swiftly and are moving towards a point of complete exhaustion.

16.12 KEY WORDS

Hazardous Wastes; recycling; incineration; landfill site; CPCB; MoEFCC; Electronic Waste; metal recovery.

Recycling: Process of collecting processing material so that they can be used again.

CPCB: Central Pollution Control Board.

Material Recovery: Obtaining from the material waste/discarded products

16.13 REFERENCES AND SUGGESTED READINGS

ASSOCHAM, 2018. A joint study by Associated Chambers of Commerce (ASSOCHAM) and NEC technologies on e-waste status of India https://in.nec.com/en_IN/pdf/ElectricalsandElectronicsManufacturinginIndia2018.pdf

CPCB, 2009. National Inventory of Hazardous Waste generating industries and Hazardous Waste Management in India. Report by CPCB

De, A.K., and De, A.K., 2013, Environmental Chemistry. New Age International Pvt. Ltd., Delhi

EU (2002). European Union Waste Electronic Electrical Waste (WEEE) Directive. Brussels: EU.

HAPPI-Home 6, 2010. Alternatives to Household Hazardous Products. Hawaii Pollution Prevention Information. College of Tropical Agriculture and Human Resources, pg. 1-2).

India: State of the Environment, 2001. Hazardous waste, with special reference to solid waste, pg. 133-149.

IRGSSA, 2005. Country level WEEE assessment study by the International Resource Group Systems South Asia Pvt. Ltd (IRGSSA), (m/s IRG Systems South Asia Pvt. Ltd).

Kudo, A. and Miyahara, S., 1991. A case History, Minamata Mercury pollution in Japan-From Loss of Human lives to decontamination. Water Science Technology, 23, 283-290.

Masters, G.M., 2001. Introduction to Environmental Science and Engineering. Prentice Hall India, Delhi.

McKinney, M.L., Schoch, R.M. and Yonavjak, L., 2007. Environmental Science: Systems and solutions, 4th Ed., Jones and Bartlett Publishers, Massachusetts, Boston.

Mercury in India: Toxic pathways, 2003. Toxics Link, pg 25.

Miller, G.T. and Spoolman, S.E., 2013. Environmental Science, 14th ed., Cengage Learning

Miller, G.T., Jr., 2007. Living in the Environment. Cengage Learning.

MOEF, 2010. Technical EIA Guidance Manual for Common Hazardous Waste Treatment, Storage and Disposal facilities. Report prepared for Ministry of Environment and Forests.

MoEF, 2016. E-waste (Management and Handling) Rules, Gazette of India, Extraordinary, Part II, Section 3, Sub-Section (i).

MoEF, 2016. Hazardous Waste (Management, Handling and Transboundary Movement) Rules, Gazette of India, Extraordinary, Part II, Section 3, Sub-Section (i).

Rajya Sabha, 2011. E-waste in India. Research Unit (LARRDIS), Rajya Sabha Secretariat, New Delhi

Toxics Link, 2007. E-waste, Factsheet. Toxics Link.

Wath, S.B., Dutt, P.S., Chakrabarti, T., 2010. E-waste scenario in India-its management and Implications. Environment Monitoring and Assessment, DOI 10.1007/s10661-010-1331-9.

WHO, 1987. Approaches to hazardous waste management, World Health Organisation. WHO/PEP/87.3

16.14 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Ans:1 There are many products of hazardous nature that we commonly use, they can be categorised as follows:

Automotive: antifreeze, brake fluid, car wax, diesel fuel, fuel oil, gasoline, kerosene, oil/gas mixtures, transmission fluid.

Cleaners: cleaning solvents, dry cleaning solutions, floor/furniture polish, oven cleaners, rust remover, silver/metal polish.

Garden Products: chemical fertilizer, fungicide, herbicide, insect repellent/fogger/killer, mouse/rat poison, pesticide etc.

Paints: deck stain/preservative, driveway sealer, epoxy, glue/adhesive, oil, latex, and lead, paint, paint stripper, solvents, thinners, turpentine, varnish.

Miscellaneous: fluorescent light bulbs, household batteries, lighter fluid, mercury, (elemental), mercury thermometers and thermostats, nail polish/remover, pool chemicals.

The wise and minimal use of these products can reduce the impact on humans and environment. The focus should be on finding suitable alternatives with natural origin that can be disposed of in an eco-friendly manner.

Ans 2: There are many problems associated with the disposal of hazardous and electronic waste. Hazardous Waste: a) During the recovery or recycling of hazardous waste, a large supply of fresh chemicals are required increasing the cost of recycling and at the same time recovered material is not of good quality; b) Incineration of waste: Incineration, although is an essential treatment method of some specific hazardous waste, the release of harmful/carcinogenic gases is unavoidable in many cases; c) disposal of waste in ground through landfills, surface impoundments and underground injection can pose threat of leakage and contamination of ground water resources in the long run.

Electronic Waste: the purified form of semiprecious and precious metals present in the electrical equipment and the integrated circuits of electronic gadgets is a major source of income for the dealers involved in recycling of waste. To extract these elements many illegal/environmentally damaging

methods are used. Open burning of wires causing air pollution; used of strong acids, being dumped in natural waters for disposal create environmental havoc. A disciplined approach towards management of these waste is the need of the hour.

